RESEARCH ARTICLE

Implications of carbon tax and energy efficiency improvement on Thai economy

Sunil Malla* and Ram M. Shrestha**

School of Environment, Resources and Development, Asian Institute of Technology
P. O. Box 4, Klong Luang, Pathumthani 12120, Thailand.

Abstract

This paper examines the economy-wide effects of carbon tax and energy efficiency improvement as policy tools to reduce emissions of greenhouse gases (GHGs) in Thailand. The country's gross domestic product (GDP) is expected to grow at faster rate than CO_2 emissions resulting in declining CO_2 emission intensity over the study period of 1998-2030. In the base case (i.e., without any carbon tax and energy efficiency improvement), the country's GDP is expected to increase by 5.7 times while the CO_2 emissions is expected increase by 3.9 times in 2030 when compared to benchmark year (1998). The corresponding CO_2 emission intensity would decrease by 33% from 1.27 kg of CO_2 per US$ in 1998 to about 0.85 kg of CO_2 emissions in 2030. Under carbon tax of US$ 200 per ton of carbon, the CO_2 emission intensity would decline by 35% by 2030 while under 1% energy efficiency improvement each year, the CO_2 emission intensity would decline by 53% by 2030 when compared to 1998. However, the loss in GDP increases with increasing carbon tax rates. The loss in GDP would be about 0.072% in 2030 with the carbon tax rate of US$ 200 per ton of carbon. Moreover, with carbon tax, there is also loss of competitiveness of domestic firms to compete with the foreign market share. The loss in net exports would be about 1.9% with tax rate of US$ 200 per ton of carbon by 2030 when compared to base case. About 18 million tons (3.1%) of the total CO_2 emissions could be reduced under US$ 200 per ton of carbon by 2030. In contrast, with energy efficiency improvement of 1% per annum, the corresponding gain in GDP and gain in net export in 2030 when compared to base case would be 1.7% and 4.7%, respectively. When compared to base case, almost 120 million tons (21%) of the total CO_2 emissions in 2030 could be reduced under the energy efficiency improvement of 1%. In terms of sectoral share in the base case, industry and transport sectors together contribute more than 86% of the total CO_2 emissions throughout the study period. Both under carbon tax and energy efficiency cases, most of the CO_2 emissions reduction would come from Industry sector. In the case of carbon tax, the range of CO_2 emissions reduction varies from 0.52% under carbon tax rate of US$ 5 per ton of carbon to about 4.5% under carbon tax rate of US$ 200 per ton of carbon by 2030. In the case of energy efficiency case, the range of CO_2 emissions reduction varies from 4.7% under energy efficiency improvement of 0.5% to about 24.4% under energy efficiency improvement of 1.0% per year by 2030. We use multi-sector small open economy dynamic computable generable equilibrium (CGE) model to address the issues.

Keywords: Computable General Equilibrium Model; Carbon Tax; Energy Efficiency; Thailand

* Sunil Malla is a Sr. Research Specialist at Asian Institute of Technology. He received Ph.D. in Economics from University of Hawaii-Manoa. He is a member of IAEE and American Economics Association (AEA).

** Ram M. Shrestha is a Professor of Energy Economics at Asian Institute of Technology. He serves as an Associate Editor of Energy Economics and Energy-The International Journal and is a member of the Editorial Advisory Board of International Energy Journal. He received The Energy Journal's Best Paper Award from International Association for Energy Economics (IAEE) in 1991.

1. INTRODUCTION

The recent international agreements such as the Kyoto Protocol to the United Nations Framework Convention on Climate Change (UNFCCC) has renewed the discussion on and revealed the need for empirical analysis on economic instruments for climate protection to guide the policy makers. One of the economic instruments as a policy tool for addressing mitigation of greenhouse gas (GHG) emissions is carbon tax. Carbon tax is a way of reducing CO_2 emissions by increasing fossil-fuel costs to encourage energy conservation through reduced demand for fossil-fuel intensive goods and services, and substitution towards more environmentally friendly technologies and processes. Moreover, carbon tax can achieve the same emission reduction target at a lower cost than conventional command-and-control regulations. Most of the studies analyzing the potential implications of carbon tax on GHG emissions and its likely economic and social impacts are limited to industrialized countries with the exception of relatively larger developing countries such as China and India.[1] In actual practice, however, only few countries in Europe have implemented carbon tax. Developing countries such as Thailand do not have the historical experience with environmental taxation and studies related to carbon tax and its implications on economy in general equilibrium framework are limited.

Rapid industrial development and growing energy demand has resulted in increasing GHG and other local air pollutant emissions in Thailand. The country does not have any GHG emission mitigation commitments presently. However, in the face of growing international concerns for mitigation of GHGs, it may be of interest for developing countries like Thailand to examine the prospect for GHG reduction. There exist only a few studies related to climate policy in the context of Thailand using the general equilibrium framework. Timilsina and Shrestha (2002, 2004) analyzed the implications of carbon tax under different alternative schemes to recycle the tax revenue and the clean development mechanism (CDM) options in Thailand in a static general equilibrium framework using 1990 social accounting matrix (SAM). Wattanakuljarus (2004) discusses the effects of tourism on social welfare through static general equilibrium model in Thailand and suggests that only one policy change can not guarantee an improvement of *ex ante* social welfare due to the presence of policy distortions even though the host country specializes in ecotourism. A study by Li (2003) describes the ancillary effect of GHG mitigation with carbon tax when local health effects are considered using the dynamic general equilibrium approach and finds that both average gross domestic production (GDP) growth as well as welfare of households would improve in Thailand. In this paper, we examine the implication of carbon tax and energy efficiency improvements in mitigating GHG emissions and its likely impact on Thai economy using a dynamic computable general equilibrium (CGE) model. The model is specifically designed to focus on the energy sector and can simulate the effects of different carbon tax structures on Thai economy over the period of 1998-2030.

The rest of the paper is divided into 7 sections. In Section 2, the present structure of Thai economy and its energy and carbon dioxide (CO_2) emissions profile are discussed. The framework of the model is presented in Section 3 and data used in this study are then discussed in Section 4. Section 5 provides a brief description of the base case and carbon tax and energy efficiency improvement scenarios. Simulation results are discussed in Section 6. The final section summarizes the key findings of the study.

[1] See for example, Whalley and Wiggle (1990), Jorgenson and Wilcoxen (1993), Aasness et al. (1996), Bovenberg and Goulder (1996), Bohringer and Rutherford (1997), Parry (1995, 1997), Parry et al. (1999), OECD (2001), Bruvoll and Larsen (2004) and Zhang and Baranzini (2004) for recent studies and debates on carbon tax in general equilibrium framework. For selected studies on carbon tax in developing countries, see Zhang (1998) for China and Bohringer et al. (2003) for India.

2. PROFILE OF THAI ECONOMY, ENERGY USE AND CO_2 EMISSIONS

Although the financial crisis of 1997 reduced the momentum of the economy, Thailand has experienced strong growth recently. The country has the second largest economy among the ASEAN[2] countries with a total GDP of US$ 143 billion in 2003 (ADB, 2004). Thai economy has strong export compared to other regions of the world. As seen from Table 1, the total value of country's export of goods and services is about US$ 98 billion while the total value of country's import is only about US$ 79 billion. The country's per capita GDP in 2000 is almost four times the figure of East Asia and more than six times the figure of South Asia (Table 1). In terms of sectoral share in total GDP, service sector accounts for the largest share (48%) followed by the industry sector (42%) in the country. The combined share (90%) of service and industry sectors is much higher compared to corresponding figures in East- and South Asia- regions but lower compared to figures in OECD and the world average. Thailand's energy consumption has been growing rapidly since 1971. The country's total energy use in 2000 is 72 million ton of oil equivalent (toe) which is more than seven times compared to 1971 when Thailand consumed just over 9.5 million toe (IEA, 2004a). This large increase has made Thailand one of the largest energy consuming countries in the Asian region. The per capita energy consumption in Thailand is 1,188 kg of oil equivalent (kgoe) which is much higher than the per capita energy consumption figures of East- and South- Asia regions. At the same time, in 2002 alone, the value of energy consumption in Thailand accounted for 14% of the GDP (about US$ 20 billion) of which the value of imported energy accounted for more than US$ 7.5 billion (EPPO, 2003).

In 2000, the country emitted about 199 million tons of CO_2 emissions from fossil fuels combustion which is more than twice its CO_2 level in 1990. Although, Thailand generally has lower CO_2 emission per capita (3.3 tons) than most of the industrialized nations and the world average (3.8 tons), the figure is much higher compared to East- and South- Asia average. On the other hand, the CO_2 emission intensity in the country (1.2 kg per 1995 US$ GDP using exchange rate) is much lower compared to corresponding figures in East- and South- Asia regions and much higher compared to corresponding figures in OECD and the world average.

3. MODEL DESCRIPTION

The model used in this paper follows the framework of Masui (2004, 2005) which emphasizes on modeling an energy sector explicitly. The model is a single country dynamic recursive CGE model. Since the recursive models are solved one period at a time, it is possible to separate the *within-period* specification from the *between-period* specification. We highlight some of the important features and issues of these two specifications in the following section.

3.1 Within-period specification

A one-period static CGE model is described for within-period specification. In this specification, the economic structure for a particular year is fully specified. The model consists of a set of "economic agents" (like consumers and producers), each of which demands and supplies commodities or services.

On the production side, inputs are categorized into four types: intermediate commodities, energy commodities, primary factors (capital and labor), and CO_2 emissions. The presence of CO_2 pollutant as a factor means that the generated CO_2 emissions should be treated as producer's costs. The model encompasses 27 sectors, including 8 energy sectors.[3] These sectors combine primary factors with intermediate- and energy- commodities to determine the level of output. Assuming perfect competition and under profit maximization, a single representative firm operates in each of the 27 sectors, and produces a single uniform commodity (good or service) using a constant return to scale (CRTS) production technology. Substitution possibilities among factors of production depend on the factors' relative prices, following a nested constant elasticity of substitution (CES) production function. Output is produced with Leontief's fixed proportion of intermediate inputs, energy inputs and primary factors composite. The primary factor composite is a Cobb-

[2] The Association of South East Asian Nations (ASEAN) includes Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand and Vietnam.
[3] Detail classification of all 27 sectors is presented in Table A.1 (Appendix A).

Douglas aggregation of labor and capital (with elasticity of substitution, $\sigma = 1$), while both the intermediate- and energy- input composites are based on Leontief aggregation ($\sigma = 0$).

On the consumption side, the consumers own the factors of production (capital and labor) and consume produced goods and services.[4] Consumers are described by one representative household. The representative household receives all income generated by providing primary factors to the production process. Factor markets are perfectly competitive and full employment of all factors is assumed. The supply of capital is fixed within a period and immobile across activities, thus implying that capital earns activity-specific return. On the other hand, labor is assumed to be mobile across activities within the country but immobile internationally. Each activity pays an activity-specific wage. A fixed share of income is saved in each time period and invested in the production sectors. The disposable income is then used for maximizing utility by purchasing commodities. At the top level, the utility of the representative household is a nested CES function of non-energy and energy final demand composite (with $\sigma = 0$). At the second level, aggregate consumption is a CES aggregation of non-energy commodities composite (with $\sigma = 1$) and the energy commodities composite (with $\sigma = 0$).

In this within-period specification, the investment (i.e., fixed capital formation) is treated as a final demand similar to the final consumption. In addition to its role as a consumer, the household also plays the role of investor by using its savings to purchase an investment good, thereby accumulating capital. An investment good is a composite of 27 Armington goods (with $\sigma = 0$).

Under the small open economy assumption, Thailand is assumed to face a perfectly elastic world demand at a fixed world price. The total domestic non-energy demand corresponds to demand of commodities and services by producers and consumers for consumption and investment. It is allocated between domestic and imported commodities following the Armington (1969) specification. That is, domestically produced and imported non-energy commodities are considered imperfect substitutes. On the other hand, for energy commodities, domestic and imported commodities are aggregated through a CES function with infinite elasticity of substitution. Moreover, all regions other than Thailand are labeled as “rest of the world” and treated as one region.

The model only considers atmospheric CO_2 emissions related to energy consumption. Taxes are imposed on CO_2 emissions and these taxes generate additional revenue. For the producers and consumers, the user cost of energy is increased by the application of the CO_2 emission tax. This cost intervenes in the producers’ selection of production factors and in the consumers’ decision about consumption categories and levels. From this perspective, CO_2 emissions can be reduced through production restructuring (substitution among fuels, and between energy and non-energy inputs) and through demand restructuring. However, since there is no distinction made between households and government in the present model, the revenue of CO_2 emission taxes are recycled as lump-sum back to the households. Energy prices are determined exogenously. In the model there are no benefits of climate policies besides those that are determined in the market.[5] To reach an equilibrium that satisfies the Walras law, the model uses an iterative process called macro-closure. In this model the ‘Investments-Savings’ (IS) global closure is chosen that imposes a strict equality between investments and savings.

3.2 Between-period specification

The static model presented in the previous section is extended to a recursive dynamic model in which selected parameters are updated based on the results from the previous periods. The main driving forces for the economic growth in this specification of the model are capital accumulation, demographic change, factor-specific productivity change and the technology improvement.

[4] For analytical simplicity, both government and household institutions are combined together as a “representative” household in the Thai economy.

[5] Some of these benefits might stem from ancillary benefits by simultaneously reducing other GHG and local air pollutants.

The process of capital accumulation is modeled endogenously, with the previous-period investment generating new capital stock for the subsequent period. In the initial year, the capital stock is estimated from the investment and growth rate in each sector. Although the allocation of new capital across sectors is influenced by each sector's initial share of aggregate capital income, the final sectoral allocation of capital in a period is dependent on the capital depreciation rate and the capital growth rate from the previous period. Furthermore, capital growth rate is dependent on both economic growth rate and labor growth rate taking into account the labor productivity change. It is assumed that the relationship between capital stock and investment to be on the optimal economic growth path.

The investment in each sector is distributed by the expected capital income in each sector. The economic growth is given exogenously, and in order to meet the given economic growth rate, the investment is estimated in advance. Total investment in each year is calculated based on investment function of acceleration principle type. In estimating the initial capital stock, not only the economic growth but also the labor input change is taken into account. Population growth is exogenously determined. It is assumed that a growing population generates a higher level of consumption demand and therefore raises the income level of household consumption. Factor-specific productivity growth is imposed exogenously on the model based on observed trends for labor and capital. On the technology side, this model distinguishes the technology embodied in the new investment from the technology embodied in the capital stock. The technology level of capital stock in the initial year is assumed to be 1. When the new capital with more efficient technology is accumulated, the average efficiency will be improved.

The model is solved as a series of equilibria, each one representing a single year. By imposing the above policy-independent dynamic adjustments, the model produces a projected or counterfactual growth path. Policy changes are then expressed in terms of changes in relevant exogenous parameters and the model is re-solved for a new series of equilibria. Numerical computation is done with general algebraic modeling system (GAMS) and its solver PATH.

4. DATA

The benchmark data used in the model comprise of three main parts: SAM, CO_2 emissions and energy prices, and scenarios. Data reported in "SAM" are value data while "CO_2 emissions and energy prices" are the quantity and value data, respectively. "Scenarios" are the percent data. The first two parts of the data set (i.e., SAM and CO_2 emissions and energy prices) are used for the within-period specification of the model and the last part of data set is used for the between-period specification of the model. All data related to the first two parts refer to 1998, which is the benchmark year for the simulation.

For value data of outputs, intermediate inputs, final consumption, taxes, investment, exports and imports, the data from original balanced SAM[6] of Thailand for the year 1998 is used which is obtained from the International Food Policy Research Institute (IFPRI). In order to use the original data set, several adjustments are made. First, the original dataset has 61 sectors, 3 primary factors, 6 institutions, 6 different tax structure, 1 investment composite, export and import. The resulting aggregated data set (hereafter modified final SAM data set) has 27 sectors, 2 primary factors, 1 institution, 5 tax structures, 1 investment composite, export and import. Within the 27 sectors, 8 are energy related sectors. At the benchmark year of the simulation, there are several taxes in Thailand. For example, there are excise tax, value added tax, special business tax, import duty, subsidy and direct tax. Based on the model specifications, we aggregated these existing taxes to five categories, namely, import tax, commodity tax, capital tax, direct tax and labor income tax. Likewise, the government, enterprises and households are aggregated into one institution.

Second, in order to make the data structure compatible with the model, further adjustments are made from the modified final SAM data set. First step is the preparation of "use matrix" (U matrix) that shows the distribution of produced commodity to activities and final demand. U matrix also records what kind and how much of input for production activity is necessary in each activity. Second step is the preparation of "make

[6] For a discussion of SAMs and their use in economy-wide policy analysis see Dervis et al. (1982), Pyatt and Robinson (1985), and Reinert and Ronand-Holst (1997).

matrix" (V matrix) that shows how much of different commodity is produced by each activity. Third and the final step is the preparation of fixed capital formation matrix that shows how much of each investment goods is bought by each sector. Since there is no investment of goods in 8 energy sectors, this data is reported for the remaining 19 sectors only. Since our main purpose is to investigate the implication of CO_2 emission tax on Thai economy, it is necessary to prepare a dataset on sectors whose input and consumption involves CO_2 emissions. Data related to energy use and their corresponding prices enable one to compute, for the 27 sectors and for the consumers, the percentage of fossil fuels consumption that generates emissions.

For dynamic model simulation, the following scenario data set are prepared: economic growth rate, growth rate by sectors, labor supply, productivity of new investment in the initial year, productivity change of labor and energy in the new investment, and the operation rate of capital. Labor productivity change is assumed to be 1 for the initial year. Differentiated labor efficiency improvement is assumed for the subsequent years for each sector. The depreciation rate for capital stock of 5% is assumed for all sectors. We assume international commodity prices are exogenous and these prices remain same during the simulation period. The GDP growth rate from year 2001 until 2016 is based on the official forecast. It is assumed that the GDP growth rate will decrease progressively from 2017 to 2020 and remains constant from 2021 to 2030. The labor growth rate data are taken from the officially forecast data of the United Nations. The operation rate of capital is assumed to remain same throughout the simulation period. In order to evaluate extraction cost, it is assumed that extraction cost increases in proportion to cumulative fuel extraction. All these parameters used in the model are summarized in Malla (2005).

5. SCENARIO DESCRIPTION

Three scenarios are considered in the study. They consist of base case scenario and two policy alternative scenarios. The base case is calibrated against the projections of CO_2 emissions, GDP and population. There is no carbon tax under the base case. This scenario maintains all policies and measures, such as the tax structures of the Thai economy, at their initial level and replicates the benchmark economy. In addition, the differentiated labor productivity improvement is assumed throughout the study period. This base case can be characterized by a high economic growth rate combined with moderate labor productivity improvement. It means that we assume a relatively low decrease in CO_2 emissions with limited changes in the fuel mix for the base case. The parameters used in the model are summarized in Table A.2 (Appendix A).

The two alternative policy scenarios are carbon tax- and energy efficiency- scenarios. Under the carbon tax scenario, six cases are considered that correspond to different carbon tax rates. Tax base starts with US$ equivalent of 5 per ton of carbon emission. Alternative taxes considered are US$ 25, 50, 100, 150 and 200 per ton of CO_2 emission.[7] We abbreviate these six tax rates under this scenario as CT5, CT25, CT50, CT100, CT150 and CT200, respectively. Carbon taxes are imposed on 8 energy sectors from year 2008 through 2030 the last period of the model.[8]

Under energy efficiency scenario, the technology level of capital stock in the initial year is assumed 1. When the new capital with more efficient technology is accumulated, the average efficiency will be improved. We assumed three cases (i.e., 0.5, 0.75 and 1.0 %) of average energy efficiency improvement alternatives throughout the simulation period. We abbreviate these three energy efficiency improvement cases under this scenario as EE0.5, EE0.75 and EE1.0. These alternative energy efficiency improvements are considered for 6 energy sectors.[9] We assume that there is no additional cost of average energy efficiency improvements.

[7] Assuming the exchange of 1 US dollar is equivalent to Thai Baht 40, the corresponding tax rates in Thai Baht are respectively, 54.5, 272.5, 545, 1090, 1635 and 2180 per ton of carbon emission. A ton of carbon corresponds to 3.67 tons of CO_2.

[8] Under Kyoto Protocol, the first commitment period is 2008 - 2012. Although Thailand does not have obligations to reduce CO_2 emissions under the protocol, we have chosen the year 2008 as a starting year of carbon tax implementation for the analysis purpose.

[9] We exclude energy efficiency improvement in two energy sectors, namely electricity- and gas distribution- sectors. The output (or goods) produced in these two sectors are electricity and town gas and these two sectors do not emit any CO_2 or other local air pollutant emissions.

6. RESULTS AND DISCUSSIONS

6.1 Base case results

Table 2 presents the estimated macro-economic variables and CO_2 emissions under the base case during the analysis period (1998 - 2030). In the base case, consumption is estimated to increase by four times (i.e., from 3 trillion Baht in 1998 to 12.5 trillion Baht in 2030), while the investment and net exports are estimated to increase by more than ten- and six- times, respectively, from 1998 to 2030. Likewise, wage rate is estimated to increase by almost three-folds from 1.0 in 1998 to 2.8 in 2030. As seen from Table 2, the total GDP is estimated to increase by 5.7 times (i.e., from 4.7 trillion Baht in 1998 to 26.9 trillion Baht in 2030). One the other hand, the CO_2 emission is expected to increase only by four times by 2030. Unlike in the past, CO_2 emission is estimated to grow less rapidly than GDP during the study period. As a result, CO_2 emission intensity is expected to decline during 1998-2030 (see Figure 1) unlike the historical trend before 1998 (IEA, 2004b). The decrease in CO_2 emission intensity is mainly due to the increase in labor productivity, the increase share of cleaner fossil-fuel (i.e., natural gas), and the change in sectoral composition from energy intensive- to non-energy intensive- sectors in the Thai economy over the analysis period. Among the components of GDP, the share of consumption and investment combined together represents about 85% of total GDP in 1998 and this share remains the same in 2030.

In terms of sectoral contribution in total CO_2 emissions, industry sector would have the highest share followed by the transport, agriculture, commercial and household sectors, respectively, during the period of 1998-2010 (Table 3). As seen from Table 3, the share of industry sector in total CO_2 emissions is, however, declining while the share of transport sector in total CO_2 emissions is increasing during the period of 2010-2030. This is mainly due to the labor productivity improvement and fuel switch in energy sub-sectors which constitute about two-thirds of total CO_2 emissions from the industry sector. The industry and transport sectors combined together would contribute about 87% of the total CO_2 emissions, while the combined share of agriculture, service and household sectors would be about 13% throughout the study period. The corresponding total CO_2 emissions from industry and transport sectors combined is about 130 million tons in 1998 and is estimated to increase to 492 million tons by 2030 (Figure 2). The result is not surprising since 73% of total final energy demand, the main source of anthropogenic CO_2 emissions, comes from these two sectors, and these two sectors together contribute about 53% of total GDP in 2003 in Thailand (DEDE, 2003). This suggests that any policies on climate protection related to energy use in Thailand should emphasize, particularly, on industry (specially, energy- and manufacturing- sub-sectors) and transport sectors in the future.

6.2 Economic impacts of carbon tax

Figure 3 summarizes the loss in GDP under different carbon tax cases over the study period. As can be seen from Figure 2, the range in GDP loss varies from 0.012 % (8 billion US$) under CT5 (i.e., US$ 5 per ton of carbon) to about 0.072 % (48.5 billion US$) under CT200 (US$ 200 per ton of carbon) in 2030. Note that the GDP loss in percentage terms during the period of 2015-2030 (under CT25, CT50, CT100, CT150 and CT200 cases) is lower than that in 2008. This higher GDP loss in percentage terms in 2008 is mainly due to the fact that the tax rates are imposed from 2008. Overtime, the loss in GDP is expected to fall compared to the loss in GDP in 2008. However, over the period of 2015-2030, the loss in GDP in percentage terms is estimated to increase with increasing carbon tax rates (Figure 3).

Table 4 summarizes the percentage change of macro-economic variables under four different carbon tax cases as compared to that in the base case for the selected future years.[10] It can also be seen from Table 4 that the effects of increasing carbon tax would increase consumption in percentage terms in 2010. Note that in the long term (i.e., year 2030), there would be a decrease in consumption in percentage term. The increase in consumption in the short run (i.e., year 2010) is mainly due to the carbon tax burden that falls primarily on the production side. In addition, the carbon tax revenue collected is recycled back to the consumption

[10] Although simulated results could be presented for each year until 2030, we have selected only two future years, i.e., 2010 and 2030.

side of the economy. In the long run, the fall in consumption in percentage term compared to short run is due to the declining effect of carbon tax overtime. The estimated results suggest that the export would fall more with increasing tax rates in 2010, just after two years of introducing the tax (i.e., in 2008) than that in year 2030. On the other hand, import would fall less in short run when compared to long run except under the tax rate of US$100 per ton of carbon. However, in the long run, both decrease in the aggregate volume of export and import would improve in year 2030. The effects of increasing carbon tax on investment would be negligible with maximum decrease of 0.001% throughout the study period.

The decrease in net export which is the main component of loss in GDP, is mainly due to the carbon tax imposed on energy- and the products of energy intensive- sectors that are traded widely in the international market. Demand for exports would fall, while import demand would rise, with import prices being relatively more attractive with the adoption of carbon tax. Thus, carbon tax is expected to lead to a reduced ability to domestic firms to compete for domestic as well foreign market share in the Thai economy. As seen from Table 4, the decrease in net export however is estimated to fall overtime from the highest of 16% (0.8 billion US$) in 2010 to about 1.9% (1.7 billion US$).

The results of the effects of different carbon tax rates on total CO_2 emissions and its sectoral share are presented in Table 5. Sectoral CO_2 emissions show that most of CO_2 emission reductions come from industry and transport sectors; on the contrary, CO_2 emissions would increase in other sectors (Table 5). However, the overall CO_2 emission reductions would increase with increasing carbon tax rates. Compared to base case, it can be seen from Table 5 that at the carbon tax rate of US$ 200 per ton of carbon, CO_2 emissions would fall by about 3.1% (6.5 million tons) in 2010, by 2.7% (10.5 million tons) in 2020, and by 3.1% (17.8 million tons) in 2030. Note that the share of total CO_2 emissions reduction in 2010 is relatively higher compared to year 2020. This is due to limited substitution possibility among the energy sources and the limited possibility of structural change in the economy as the tax would be imposed only from year 2008. Over the long run, CO_2 emissions reduction potential would be larger (Table 5).

Existing studies show that a carbon tax could have either positive or negative effects on GDP depending mainly on the way tax is revenue is recycled. Jorgenson and Wilcoxen (1993) shows that a carbon tax for reducing 32% of CO_2 emissions in the U.S. in year 2020 would cause a reduction in real gross national product (GNP) by 1.7% if the tax is recycled to households through a lump-sum transfer. The GNP would, however, be reduced by 0.7% if the tax revenue is used to finance cuts in labor tax rates, while it would increase by 1.1% if the tax revenue is used to finance cuts in the existing capital tax rates. In the case of China, Zhang (2000) finds that there would be a 1.47% reduction in GNP and a 20% reduction in CO_2 emissions in 2010 when the tax revenue is recycled to finance cuts in indirect taxes. Bahn and Frei (2000) finds a 0.01% decrease in GDP and a 10% reduction in CO_2 emissions in 2010 in Switzerland when the tax revenue is used to reduce social security charges. In the case Norway, a study by Bruvoll and Larsen (2004) finds that the average carbon tax of US$21 per ton of CO_2 emissions contributed to a reduction in emissions of 2.3 %. In the case of Thailand, if the CO_2 emissions were to reduce by 3%, the corresponding GDP loss would be 0.08% in 2010 and 0.072% in 2030. This suggest that the effect of carbon tax on GDP loss in Thailand would be relatively small compared to China but the effect would be relatively larger compared to industrialized countries.

6.3 Economic impacts of energy efficiency improvement

Table 6 summarizes the change in macro-economic variables under three alternative energy efficiency improvement cases for selected future years in Thailand. As can be seen from Table 6, all macro-economic variables would improve with increasing energy efficiency improvement over the study period. Consumption and import, in particular, would improve over the study period. Compared to base case, consumption would increase from 0.34% under EE0.5 (i.e., energy efficiency improvement of 0.5%) in 2010 to about 2.0% under EE1.0 (i.e., energy efficiency improvement of 1.0%) in 2030, while import volume would decrease from 0.17% under EE0.5 in 2010 and to about 1.07% under EE1.0 in 2030 (Table 6). The improvements in investment and export would be, however, small ranging from less than 0.001% under EE0.5 to about 0.24% under EE1.0 in 2030. Overall, there would be a gain in GDP from 0.23% under EE0.5 in 2010 to about 1.6% under EE1.0 in 2030.

The total CO_2 emissions reduction would increase with increasing energy efficiency improvements over the study period. As seen from Table 6, the total CO_2 emissions reduction would be about 1.1% (2.3 million tons) under EE0.5 in 2010 to about 20.6 (118 million tons) under EE1.0 in 2030. As can be seen from Figure 4, more than two-thirds of total CO_2 emissions reduction, in percentage terms, would come from combined industry and transport sectors alone. Among the sectors, the share of industry sector in total CO_2 emissions reduction would be the largest followed by both transport and other sectors. The share of industry sector would range from 1.4% under EE0.5 in 2010 to about 24.4% under EE1.0 in 2030. The corresponding figure for transport sector would from 0.9% to 17.7%, and for others sector would be from 0.9% to 18.7%.

CO_2 emission intensity is expected to decline both under carbon tax and energy efficiency improvement cases over the study period. It can be seen from Table 7 that by 2030, CO_2 emission intensity would decline by about 16% compared to benchmark year (1998) under all four carbon tax rates. Under carbon tax cases, the average annual decline rate (AADR) during the period 1998-2030 varies from 0.57% under CT5 to about 0.56% under CT200. On the other hand, the decrease in CO_2 emission intensity under the energy efficiency improvement cases would be much larger than that in the carbon tax cases. Almost 25% of CO_2 emission intensity would be reduced under EE0.5 (with an AADR of 1.62%) while about 37% of CO_2 emission intensity would be reduced under EE1.0 in 2030. The corresponding figure in the case of carbon tax rates of CT5 and CT200 would be 0.57% and 0.56%, respectively.

7. CONCLUSIONS

This paper assesses the implications of carbon tax rates and energy efficiency improvement on the Thai economy using the recursive dynamic CGE model. In the base case, the study shows that the growth rate of GDP would be relatively larger than the growth rate of CO_2 emission over the study period. The total GDP is expected to increase by 5.7 times while the CO_2 emission is expected to increase by only 3.9 times in 2030. This difference in growth rates of GDP and CO_2 is mainly due to both energy efficiency and labor productivity improvements in the Thai economy. The study shows that the annual loss in GDP would be relatively small, about 0.072% even with the carbon tax rate of US$ 200 per ton of carbon. Also, of equal importance is the loss of competitiveness of domestic firms in the export market. The CO_2 emission intensity in Thailand is expected to decrease to about 0.85 under carbon tax rate of US$ 5 per ton of carbon and to about 0.83 under US$ 200 per ton of carbon by 2030. Compared to CO_2 emission intensity in Japan (0.37 kg of CO_2 per 1995 US$) and OECD average (0.51 kg of CO_2 per 1995 US$), the corresponding figure in 2030 in Thailand under different tax rates (in the range of 0.8 – 0.83 kg of CO_2 per US$) and under different energy efficiency cases (in the range of 0.75 – 0.67 kg of CO_2 per US$) would still be higher.

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Table 1. GDP, energy and CO_2 emissions profile of Thailand and selected world regions (2000)

	Thailand	East Asia[§]	South Asia[§]	OECD[§]	World
GDP (constant 1995 billion US$ using exchange rate)	172	1,541	622	26,835	34,217
GDP per capita (constant 1995 US$)	2,828	883	460	29,866	5,654
CO_2 emissions (million tons)	199	3,553	1,220	12,449[†]	22,995
CO_2 emissions per capita (tons)	3.3	2.0	0.9	11.1[†]	3.8
CO_2 emissions intensity (kg per 1995 US$ of GDP)	1.2	2.3	2.0	0.4	0.7
Energy use (million tons of oil equivalent)	72	1,467	631	5,316[†]	9,935
Energy use (kg of oil equivalent per capita)	1,188	856	469	4,738[†]	1,694
% share macro-economic components of GDP					
Consumption	64	62	77	77	77
Export	57	39	15	24	27
Import	-46	-33	-16	-24	-27
Investment	25	32	24	23	23
% sectoral share of GDP					
Agriculture	10	16	25	2	4
Industry	42	48	27	29	31
Service	48	36	49	68	65

[§] Definition of the world regions are based on WDI (2004). East Asia excludes Thailand.
[†] These figures are based on IEA (2004a). Source: WDI (2004)

Table 2. Estimated macro-economic variables and CO_2 emissions under base case for selected future years

	1998	2010	2015	2020	2025	2030	% change (1998-2030)
	(in trillion Baht unless otherwise stated)						
Consumption	3.0	3.6	5.4	9.0	10.8	12.5	314
Investment	1.0	5.1	7.0	7.4	8.6	10.6	1007
Export	2.7	4.9	7.1	10.1	13.0	16.6	513
Import	2.0	4.7	6.6	8.7	10.5	12.9	539
Fixed capital formation	21.2	40.9	58.0	78.6	95.9	116.8	450
Labor income	1.4	1.6	1.7	1.7	1.8	1.9	30
Capital income	2.6	5.6	8.5	12.1	14.7	17.9	589
GDP	4.7	9.0	12.8	17.9	21.9	26.9	474
Wage rate (1998=1)	1.0	1.5	2.0	2.5	2.5	2.8	178
CO_2 emissions (million tons)	147	210	282	386	469	573	286

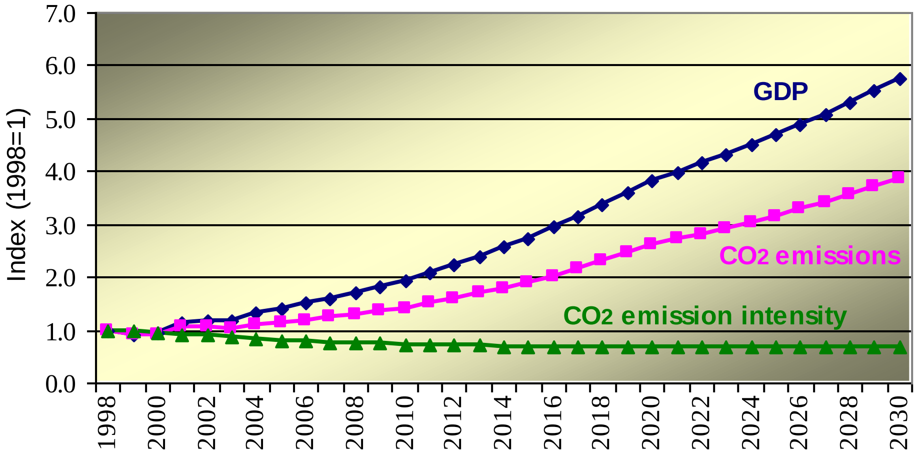
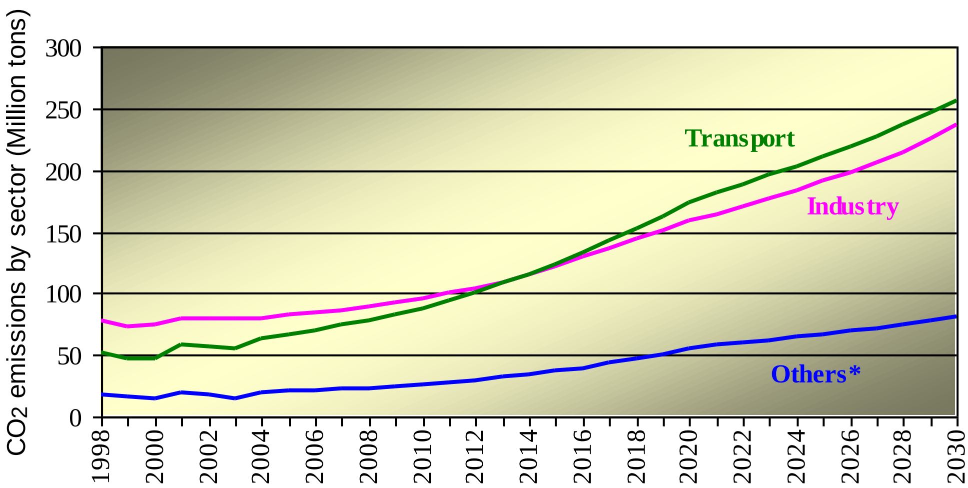


Figure 1.GDP, CO_2 emissions and CO_2 emission intensity index in Thailand under base case over the study period (1998=1)

Table 3. Estimated sectoral share of total CO_2 emissions under the base case for 1998 and selected future years (%)

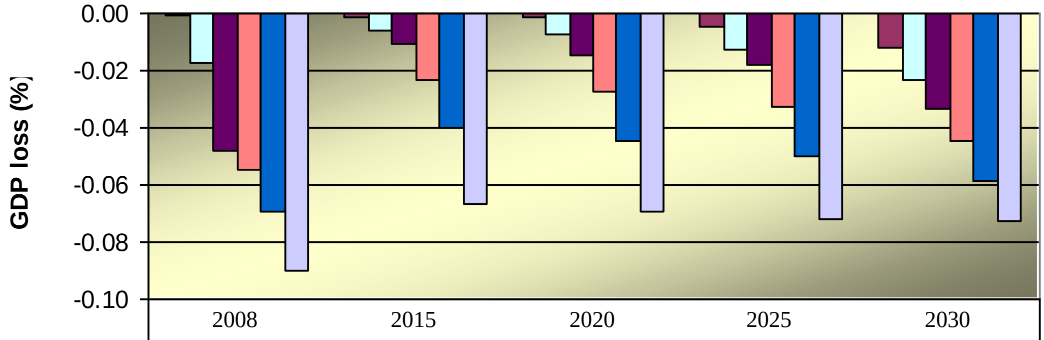
	1998	2005	2010	2015	2020	2025	2030
Agriculture	9.0	9.1	9.4	9.9	10.6	10.7	10.6
Commercial	1.9	1.8	1.8	1.9	2.1	2.1	2.1
Household	1.5	1.2	1.2	1.3	1.5	1.5	1.4
Industry	52.4	48.5	45.9	43.1	41.0	40.7	40.9
Transport	35.2	39.4	41.7	43.8	44.8	45.0	45.0
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Note: Among 27 sectors (see Table A.1 in Appendix A), there are 23 sectors under industry, 1 sector under agriculture (AGR), 1 sector under transport (TRA), and 2 sectors under commercial (HRT and SVR). These 23 industry sectors are 8 energy sub-sectors (COL, CRU, GOL, DSL, AVF, FUO, ELE and GAS), 13 manufacturing sub-sectors (OMN, FBT, TAL, TWP, PPP, CRP, PRP, NMM, MEP, MAC, TEP, OMF and CNS) and 2 other sub-sectors (WAT and TRD).



* Others include agriculture, commercial and household sectors.

Figure 2: CO_2 emissions by sectors during the study period under base case (million tons)



	2008	2015	2020	2025	2030
CT5	0.000	-0.001	-0.002	-0.005	-0.012
CT25	-0.017	-0.006	-0.007	-0.013	-0.023
CT50	-0.048	-0.011	-0.014	-0.018	-0.033
CT100	-0.055	-0.023	-0.027	-0.033	-0.045
CT150	-0.069	-0.040	-0.045	-0.050	-0.059
CT200	-0.090	-0.067	-0.069	-0.072	-0.072

Figure 3. Loss in GDP under different carbon tax rates compared to base case (%).

Table 4. Estimated effects of different carbon tax cases on selected macro economic variables compared to base case (%)

	2010				2030			
	CT5	CT25	CT100	CT200	CT5	CT25	CT100	CT200
Consumption	0.002	0.105	0.410	0.724	0.059	0.233	0.263	0.401
Investment	-0.001	-0.004	-0.016	-0.029	<-0.001	<0.001	0.001	0.002
Export	<-0.001	-0.191	-0.590	-0.924	-0.100	-0.261	-0.383	-0.640
Import	-0.002	-0.083	-0.211	-0.272	-0.046	-0.061	-0.145	-0.284
Net export*	<-0.000	-2.701	-9.389	-16.065	-0.117	-0.943	-1.196	-1.860

* Net export is the difference of export and import.

Table 5. Estimated effects of different carbon tax rates on sectoral CO_2 emissions for selected future years (%)

	2010				2020				2030			
	CT5	CT25	CT100	CT200	CT5	CT25	CT100	CT200	CT5	CT25	CT100	CT200
Others*	0.00	0.07	0.25	0.42	0.00	0.01	0.07	0.27	0.03	0.14	0.21	0.33
Industry	-0.01	-0.83	-3.09	-4.84	-0.02	-0.09	-0.85	-3.36	-0.52	-2.11	-2.96	-4.52
Transport	-0.07	-0.30	-1.16	-2.26	-0.09	-0.43	-1.59	-2.92	-0.06	-0.32	-1.48	-2.84
Total	-0.03	-0.50	-1.87	-3.11	-0.05	-0.23	-1.05	-2.65	-0.24	-0.99	-1.85	-3.09

* Others include agriculture, service and household sectors.

Table 6. Estimated effects of different energy efficiency improvement cases on macro economic variables and CO_2 emissions for selected future years (%)

	2010			2020			2030		
	EE0.5	EE0.75	EE1.0	EE0.5	EE0.75	EE1.0	EE0.5	EE0.75	EE1.0
Consumption	0.34	0.50	0.66	0.69	0.93	0.99	1.02	1.53	2.00
Investment	0.01	0.01	0.01	0.02	0.02	0.02	0.02	0.02	0.02
Export	<0.00	<0.00	<0.00	<0.00	0.16	0.46	0.24	0.24	0.24
Import	-0.17	-0.24	-0.32	-0.39	-0.52	-0.65	-0.56	-0.83	-1.07
GDP	0.23	0.34	0.44	0.55	0.81	1.08	0.89	1.26	1.60
CO_2 emissions	1.12	4.08	5.37	2.95	10.24	12.70	4.26	15.93	20.64

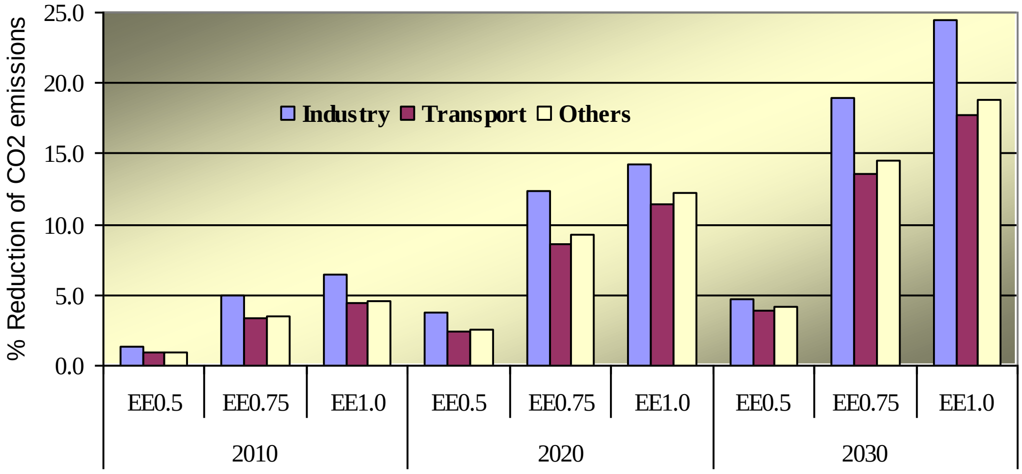


Figure 4. Percentage reduction of CO_2 emissions by sectors compared to base case under different energy efficiency improvement cases for selected future years (%)

Table 7. Estimated CO_2 emission intensity under different carbon tax and energy efficiency cases for selected future years (kg of CO_2 per US$)

	Carbon tax				Energy efficiency		
	CT5	CT25	CT100	CT200	EE0.5	EE0.75	EE1.0
2010	0.93	0.93	0.92	0.91	0.91	0.89	0.88
2015	0.88	0.88	0.87	0.86	0.84	0.82	0.80
2020	0.86	0.86	0.86	0.84	0.80	0.77	0.75
2025	0.85	0.85	0.85	0.83	0.78	0.74	0.71
2030	0.85	0.84	0.84	0.83	0.75	0.71	0.67
AADR*	0.57	0.58	0.56	0.56	1.62	1.81	1.99

* AADR is average annual decline rate.

APPENDIX A

Table A.1: Sectors/commodities classification considered in the study

S. No.	Sector/Commodity	Code
1	Agriculture, livestock, forestry and fishery	AGR
2	Coal and lignite	COL
3	Crude petrol and natural gas	CRU
4	Other mining	OMN
5	Food, beverage and tobacco	FBT
6	Textile, leather, and the products	TAL
7	Timber and wooden products	TWP
8	Pulp, paper and printing	PPP
9	Chemical products	CRP
10	Gasoline	GOL
11	Diesel	DSL
12	Aviation fuel	AVF
13	Fuel oil	FUO
14	Plastic and rubber products	PRP
15	Non-metallic mineral products	NMM
16	Metal products	MEP
17	Machinery	MAC
18	Transport equipment	TEP
19	Other manufacturing products	OMF
20	Electricity	ELE
21	Gas distribution	GAS
22	Water	WAT
23	Construction	CNS
24	Trade	TRD
25	Hotels and restaurants	HRT
26	Transport	TRA
27	Services	SVR

Table A.2: Growth rates of GDP, population and labor productivity improvements (%), and capital operation rate assumed in the model during the study period

Year	Growth rates (%)			Operation rate
	GDP	Population	Labor productivity	
1998	-2.24	0.53	2.00	0.80
1999	0.23	0.31	2.00	0.80
2000	6.03	0.71	2.10	0.85
2001	2.67	1.47	2.20	0.90
2002	5.16	1.23	2.30	0.95
2003	8.71	1.21	2.40	1.00
2004	6.43	1.20	2.50	1.00
2005	6.65	1.13	2.60	1.00
2006	6.72	1.08	2.70	1.00
2007	6.80	1.04	2.80	1.00
2008	7.35	1.00	2.90	1.00
2009	7.47	0.97	3.00	1.00
2010	7.59	0.93	3.10	1.00
2011	7.80	0.90	3.20	1.00
2012	7.90	0.88	3.30	1.00
2013	7.52	0.85	3.40	1.00
2014	7.66	0.83	3.50	1.00
2015	7.97	0.81	3.50	1.00
2016	8.09	0.80	3.40	1.00
2017	7.48	0.78	3.30	1.00
2018	6.86	0.76	3.20	1.00
2019	6.24	0.74	3.10	1.00
2020	5.62	0.72	3.00	1.00
2021	5.00	0.70	2.90	1.00
2022	5.00	0.68	2.80	1.00
2023	5.00	0.66	2.70	1.00
2024	5.00	0.64	2.60	1.00
2025	5.00	0.62	2.50	1.00
2026	5.00	0.60	2.40	1.00
2027	5.00	0.57	2.30	1.00
2028	5.00	0.55	2.20	1.00
2029	5.00	0.52	2.10	1.00
2030	5.00	0.50	2.00	1.00

Note: GDP growth rate from 1998 to 2016 are based on TDRI (2004). The population growth rate from 1998 to 2030 is based on United Nations (2005). The labor productivity growth rate and operation rate are assumed based on authors' judgment.

APPENDIX B

In this section we describe the simplified version of the mathematical formulation used in the study. Imagine a small open economy that represents different sectors, commodities, agents, and institutions that recognize feedbacks and inter-relationships between sectors and allow all economic agents within all sectors to optimize. Although the model distinguishes between energy and non-energy sectors or commodities, for easy mathematical exposition, we do not distinguish between energy and non-energy sectors. Let *t* denote time period, *i* (*i* = 1,2,…,m)and *j* (*j* = 1,2,…,n) denote commodity and sector index, respectively. Moreover, we define the following notations as: *Y* = produced commodities, *X* = intermediate inputs, *C* = final consumption, *I* = investment, *P* = price level, and *S* = savings. The market equilibrium for produced commodities can then be represented by the following equations:

$$\begin{aligned}
&P_{ti}\left\{\sum_{j=1}^{n} Y_{tji} - \left(\sum_{j=1}^{n} X_{tij} + C_{ti} + \sum_{j=1}^{n} I_{tij}\right)\right\} = 0, \\
&P_{ti} \geq 0 \quad and \\
&\sum_{j=1}^{n} Y_{tji} - \left(\sum_{j=1}^{n} X_{tij} + C_{ti} + \sum_{j=1}^{n} I_{tij}\right) \geq 0
\end{aligned} \tag{B.1}$$

These equations indicate the relationship between the commodity price P_{ti} and the demand and supply of that commodity. These equations state that for equilibrium to exist, either the price (P_{ti}) has to be zero (excess supply) or the excess demand for each commodity *i* has to be zero. The important insight here is that for equilibrium to exist in the goods market, supply must be equal to demand in all goods market.

Likewise, market for production factors (capital and labor) can be represented by the following equations:

$$\begin{aligned}
&P_{tk}\left\{K_t^* - \sum_{j=1}^{n} K_{tj}\right\} = 0, \\
&P_{tk} \geq 0 \quad and \\
&K_t^* - \sum_{j=1}^{n} K_{tj} \geq 0
\end{aligned} \tag{B.2}$$

and

$$\begin{aligned}
&P_{tl}\left\{L_t^* - \sum_{j=1}^{n} L_{tj}\right\} = 0 \\
&P_{tl} \geq 0 \quad and \\
&L_t^* - \sum_{j=1}^{n} L_{tj} \geq 0
\end{aligned} \tag{B.3}$$

In the capital market and the labor market (Equations B.2 and B.3), total demand for these is less than the endowment of capital and labor, respectively. In a state of equilibrium for these production factors, these prices become positive values, but in a state of disequilibrium, that is, when the supply of these factors is more than the demand, the prices become 0.

In order to introduce CO_2 emission in the model, the new commodity, E, is introduced. This is similar in nature as that of CO_2 emission right. It is assumed that when carbon tax is introduced, this emission right would also be bought when the fuel is demanded. For simplicity, we assume the supplier of this emission right is households. It can be seen from Equations B.4 that if the demand of emission right exceeds supply, the price of the emission right will be positive. On the other hand, if the supply exceeds the demand, the price is 0. The price of this emission right is equivalent to the level of carbon tax to reduce CO_2 emissions.

$$P_{te}\left\{E_t^* - \sum_{j=1}^{n} E_{tj}\right\} = 0$$
$$P_{te} \geq 0 \quad and \tag{B.4}$$
$$E_t^* - \sum_{j=1}^{n} E_{tj} \geq 0$$

On the production side, the revenues and expenditures in each production sectors can be expressed by the following equation:

$$\sum_{i=1}^{m} P_{ti} X_{tij} + P_{tk} K_{tj} + P_{tl} L_{tj} + P_{te} E_{tj} = \sum_{i=1}^{m} p_{ti} Y_{tij} \tag{B.5}$$

Expenditure for buying intermediate goods and other production factors in order to produce commodities is represented by left hand side while the right-hand side represents the total revenues from selling the produced commodities in each sector *j*. The Equation B.5 also shows that the revenues and expenditures are balanced in sector *j*. The relationship between inputs and output can be represented by the following production function:

$$Y_{tij} = \min\left(\frac{X_{tij}}{m_{ij}}, \frac{\alpha_{0j} K_{tj}^{1-\alpha_{1j}}}{g_j}\right) \tag{B.6}$$

where α_{0j} and α_{1j} are adjustment parameter in sector *j* and the share of capital cost in sector *j*, respectively, and m_{ij} and g_j are input coefficients in sector *j*.

On the consumption side, the income of households includes not only from renting their labor and capital to the production sectors, but also receives the revenue collected from the carbon tax. The income of the household represented by Equation B.7.

$$H_t = P_{tk} \sum_{j=1}^{n} K_{tj} + P_{tl} \sum_{j=1}^{n} L_{ij} + P_{te} \sum_{j=1}^{n} E_{tj} \tag{B.7}$$

The demand for each commodity is obtained by maximizing the household utility subject the income constraints. Here, savings, S_t, is included in the demand function as the demand for investment goods I_{tij} besides the final consumption C_{ti}. The next two equations (Equations B.8 and B.9) show the expenditure of the household sector, and the demand function where U_t is the utility, β_0 is the adjustment parameter, β_{1i} is the monetary share of final consumption goods *i* to total income.

$$H_t = \sum_{i=1}^{m} P_{ti}\left(C_{ti} + \sum_{j=1}^{n} I_{tij}\right), \tag{B.8}$$

$$U_t = \beta_0 C_{ti}^{\beta_{1i}} S_t^{1-\sum_{i=1}^{m}\beta_{1i}} \quad and \quad S_t = \sum_{i=1}^{m}\sum_{j=1}^{n} I_{tij}. \tag{B.9}$$

Savings S_t are distributed in investment in each sector. The share of investment in each sector is calculated by assuming the rate of return from capital stock in each sector. The contents of investment goods in each sector, I_{tij}, are calculated in proportion to the total investment I_{tj} as follows:

$$I_{tij} = \gamma_{ij} I_{tj}, \quad \sum_{i=1}^{m} \gamma_{ij} = 1. \tag{B.10}$$

Where γ_{ij} is the parameter of investment goods contents in sector j. The total new investment is added to the existing stock depleted at δ given by the Equation B.11:

$$K_{t+1,j} = (1-\delta_j)K_{tj} + I_{tj} \tag{B.11}$$

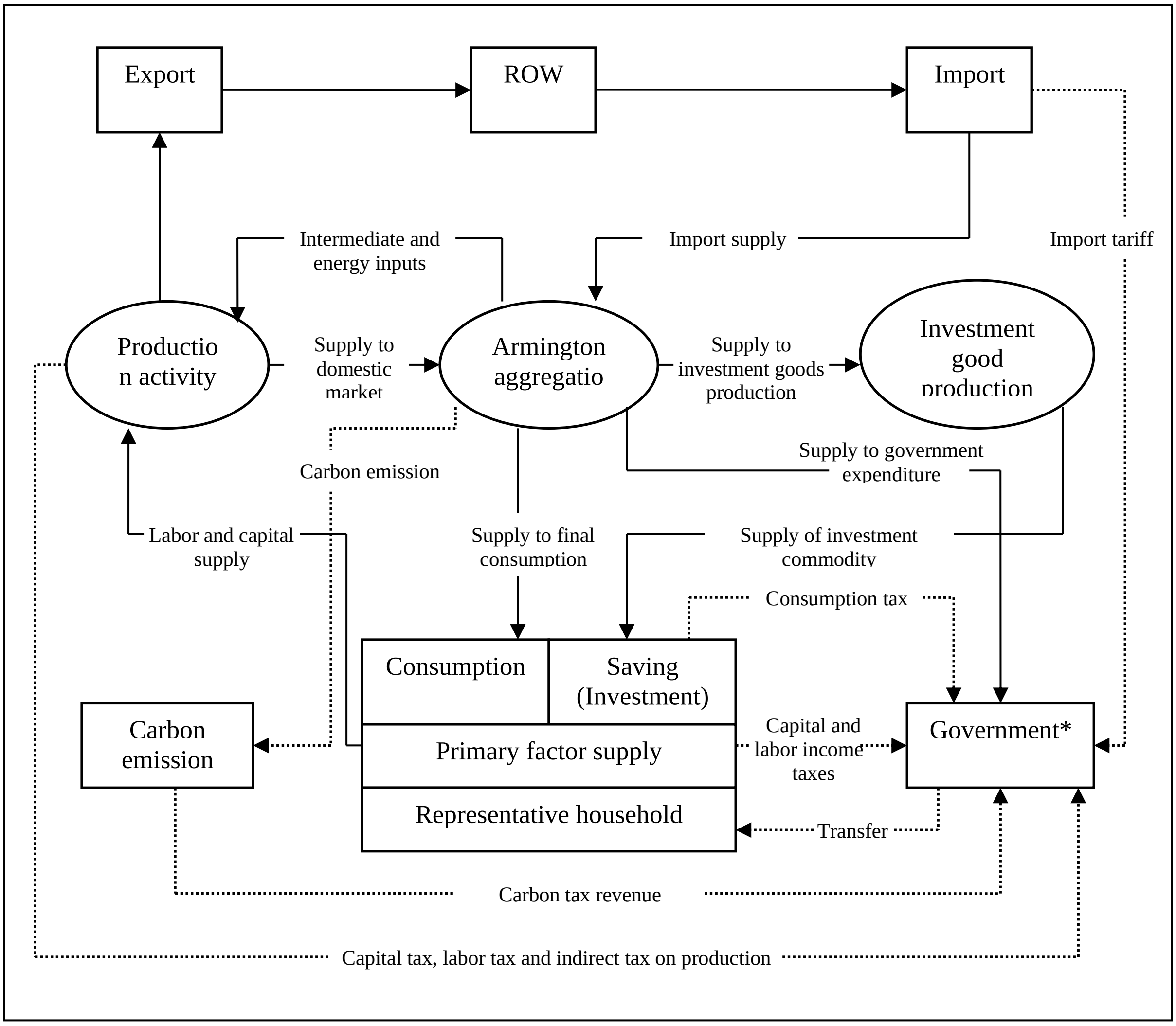
The parameter of technology improvement, b_{tjx} (where x is the index of efficiency such as labor efficiency and energy efficiency), is updated by using the efficiency of the previous year, b_{t-1jx}, and the new technology efficiency embodied in new investment, b_{ntjx}, is given by the following Equation B.12:

$$b_{tjx} = \frac{b_{t-1,jx}(1-\delta_j)K_{t-1,j} + bn_{tjx} I_{tj}}{K_{tj}} \tag{B.12}$$

The capital stock in the initial year is calculated based on the following Equation B.13:

$$K_t = \frac{I_t}{\delta + \left[\frac{1+g}{(1+l)^{\alpha_L}}\right]^{\frac{1}{\alpha_K}} - 1} \tag{B.13}$$

where l and g are labor growth rate (taking into account the labor productivity change) and economic growth rate, respectively, and α_k and α_L are the capital and labor share, respectively. Using this formulation, the capital stock is calculated in the model.



*Note: In our simplified model in the paper, we do not represent government as a separate institution. In order words, all the flow in and out of government block is combined to representative household.

Figure B.1. Flow of goods, factors and taxes of the model.